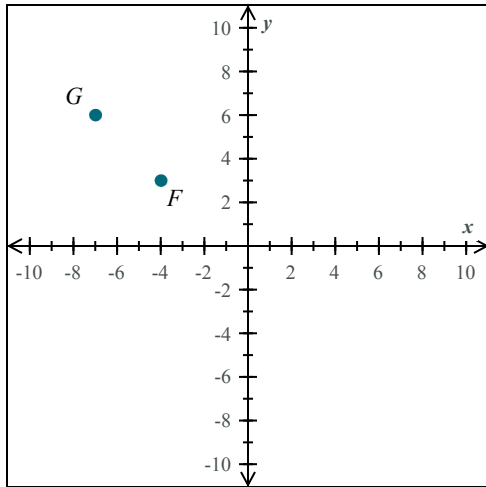


Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **25**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 25**

Calculate the distance between the points $G = (-7, 6)$ and $F = (-4, 3)$ in the coordinate plane.

Give an exact answer (not a decimal approximation).



Distance: _____

Question 2 of 25

Find an equation for the line that passes through the points $(-3, -4)$ and $(1, 6)$.

Question 3 of 25

Find the y -intercept(s) and x -intercept(s) of the graph of the following.

$$y = \frac{9x + 5}{x^2 + 7}$$

If there is more than one answer, separate them with commas.

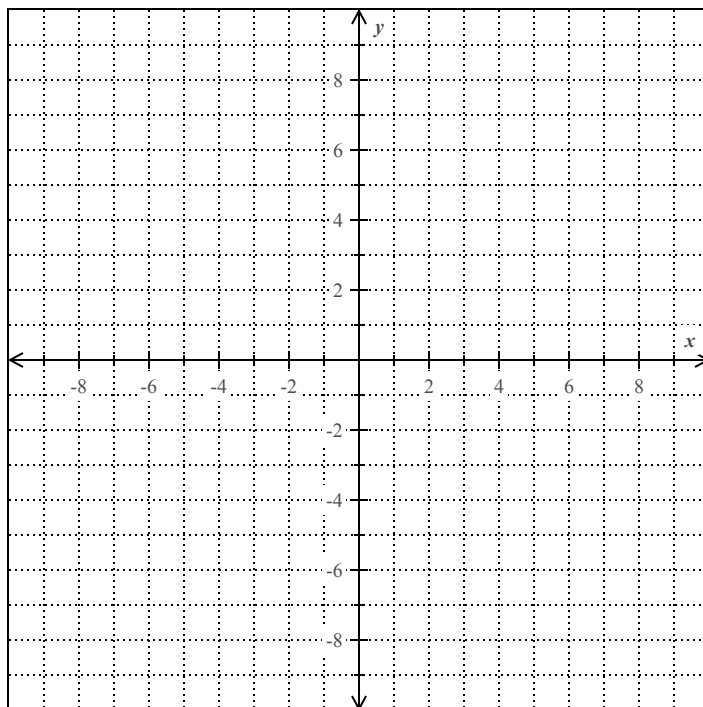
Question 4 of 25

The equation of a circle is given below. Identify the center and radius. Then graph the circle.

$$(x - 1)^2 + (y - 3)^2 = 4$$

Center: (_____, _____)

Radius: _____



Question 5 of 25

The functions f and g are defined as follows.

$$f(x) = \frac{x-3}{x^2+6x+9}$$

$$g(x) = \frac{x-6}{x^2-5x-6}$$

For each function, find the domain.

Write each answer as an interval or union of intervals.

Domain of f : _____

Domain of g : _____

Question 6 of 25

The functions f , g , and h are defined as follows.

$$f(x) = \sqrt{-2-2x} \quad g(x) = \left| -10 + \frac{1}{2}x \right| \quad h(x) = \frac{11+x^3}{x^3}$$

Find $f(-3)$, $g(8)$, and $h(-3)$.

Simplify your answers as much as possible.

$$f(-3) = \underline{\hspace{2cm}}$$

$$g(8) = \underline{\hspace{2cm}}$$

$$h(-3) = \underline{\hspace{2cm}}$$

Question 7 of 25

Suppose that the function h is defined on the interval $(-2, 2]$ as follows.

$$h(x) = \begin{cases} -1 & \text{if } -2 < x \leq -1 \\ 0 & \text{if } -1 < x \leq 0 \\ 1 & \text{if } 0 < x \leq 1 \\ 2 & \text{if } 1 < x \leq 2 \end{cases}$$

Find $h(-1)$, $h(0.5)$, and $h(2)$.

$h(-1) =$	
$h(0.5) =$	
$h(2) =$	

Question 8 of 25

Consider the line $y = 4x - 7$.

- (a) Find the equation of the line that is parallel to this line and passes through the point $(6, 6)$.
- (b) Find the equation of the line that is perpendicular to this line and passes through the point $(6, 6)$.

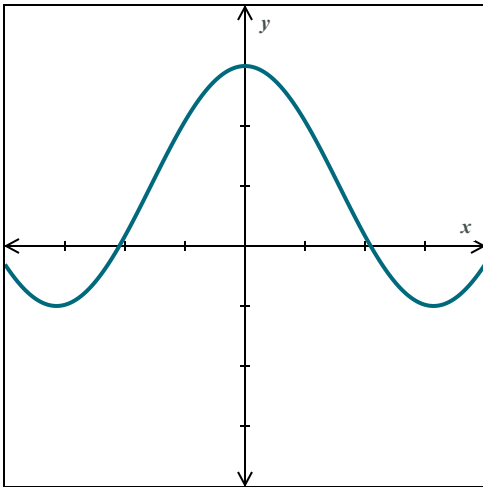
Note that a graphing calculator may be helpful in checking your answer.

Question 9 of 25

Four functions are given below. Either the function is defined explicitly, or the entire graph of the function is shown.

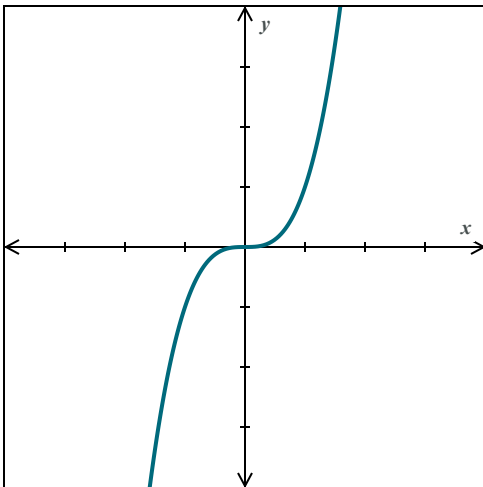
For each, decide whether it is an even function, an odd function, or neither.

The function r



- Even - Odd - Neither

The function s



- Even - Odd - Neither

$$g(x) = -6x^5 + 7x^2$$

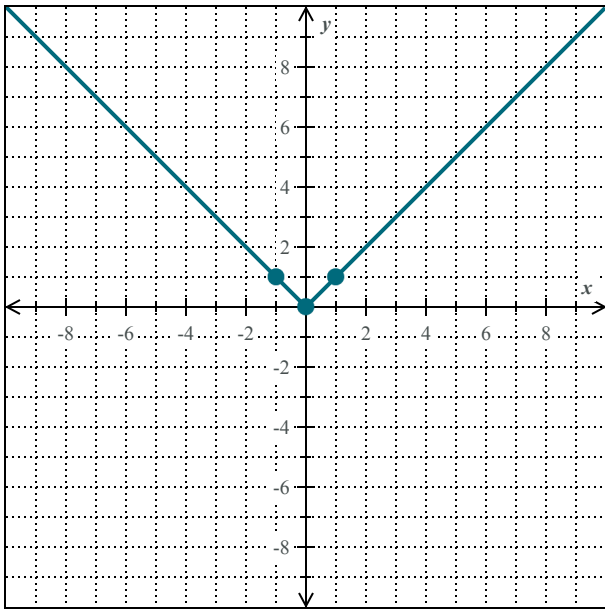
- Even - Odd - Neither

$$h(x) = -4x^5 + 7x^3$$

- Even - Odd - Neither

Question 10 of 25

Below is the graph of $y = |x|$. Translate it to make it the graph of $y = |x + 4| + 2$.



Question 11 of 25

The function h is defined as follows.

$$h(x) = x^2 - 3$$

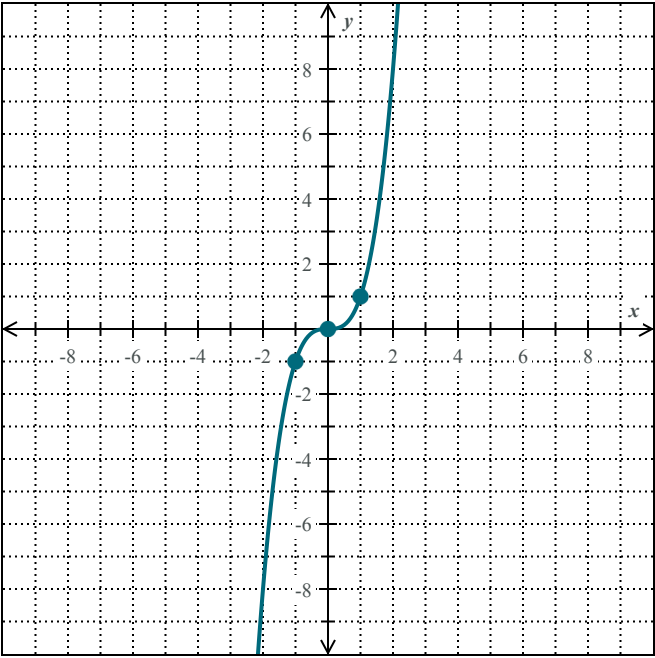
If the graph of h is translated vertically upward by 8 units, it becomes the graph of a function g .

Find the expression for $g(x)$.

$$g(x) = \boxed{}$$

Question 12 of 25

Below is the graph of $y = x^3$. Translate it to make it the graph of $y = (x - 1)^3 - 3$.



Question 13 of 25

Find the difference quotient $\frac{f(x + h) - f(x)}{h}$, where $h \neq 0$, for the function below.

$f(x) = 2x - 2$

Simplify your answer as much as possible.

$$\frac{f(x + h) - f(x)}{h} =$$

Question 14 of 25

Goran and his wife are each starting a saving plan. Goran will initially set aside \$650 and then add \$155 every month to the savings. The amount A (in dollars) saved this way is given by the function $A = 650 + 155N$, where N is the number of months he has been saving.

His wife will not set an initial amount aside but will add \$485 to the savings every month. The amount B (in dollars) saved using this plan is given by the function $B = 485N$.

Let T be total amount (in dollars) saved using both plans combined. Write an equation relating T to N . Simplify your answer as much as possible.

Question 15 of 25

Suppose that the functions g and h are defined for all real numbers x as follows.

$$g(x) = 4x^3$$

$$h(x) = x^2$$

Find the following.

$$(g + h)(x) = \underline{\hspace{4cm}}$$

$$(g \cdot h)(x) = \underline{\hspace{3cm}}$$

$$(g - h)(2) = \underline{\hspace{1.5cm}}$$

Question 16 of 25

For the real-valued functions $f(x) = \frac{x-2}{x-1}$ and $g(x) = 4x + 9$, find the composition $f \circ g$. Also, specify its domain using interval notation.

$(f \circ g)(x) =$

Domain of $f \circ g$:

For the real-valued functions $f(x) = \frac{x-2}{x-1}$ and $g(x) = 4x + 9$, find the composition $f \circ g$. Also, specify its domain using interval notation.

$f \circ g \ x =$

Domain of $f \circ g$:

Question 17 of 25

Suppose $H(x) = (8 - 3x)^2$.

Find two functions f and g such that $(f \circ g)(x) = H(x)$.

Neither function can be the identity function.
(There may be more than one correct answer.)

$f(x) =$ _____

$g(x) =$ _____

Question 18 of 25

The volume $V(r)$ (in cubic meters) of a spherical balloon with radius r meters is given by

$$V(r) = \frac{4}{3} \pi r^3.$$

The radius $W(t)$ (in meters) after t seconds is given by $W(t) = 7t + 3$.

Write a formula for the volume $M(t)$ (in cubic meters) of the balloon after t seconds.

It is not necessary to simplify.

$$M(t) =$$

Question 19 of 25

For the polynomial below, 3 is a zero.

$$g(x) = x^3 - 9x^2 + 20x - 6$$

Express $g(x)$ as a product of linear factors.

Question 20 of 25

Consider the following polynomial function.

$$f(x) = -(x + 1)^2(x - 1)$$

Answer the questions regarding the graph of f .
Then, use this information to graph the function.

(a) Choose the end behavior of the graph of f .

(Blank 1)

Blank 1 Options

- Falls to the left and rises to the right
- Rises to the left and falls to the right
- Rises to the left and rises to the right
- Falls to the left and falls to the right

(b) List each real zero of f according to the behavior of the graph at the x -axis near that zero. If there is more than one answer, separate them with commas. If there is no answer, write "None".

Zero(s) where the graph crosses the x -axis:

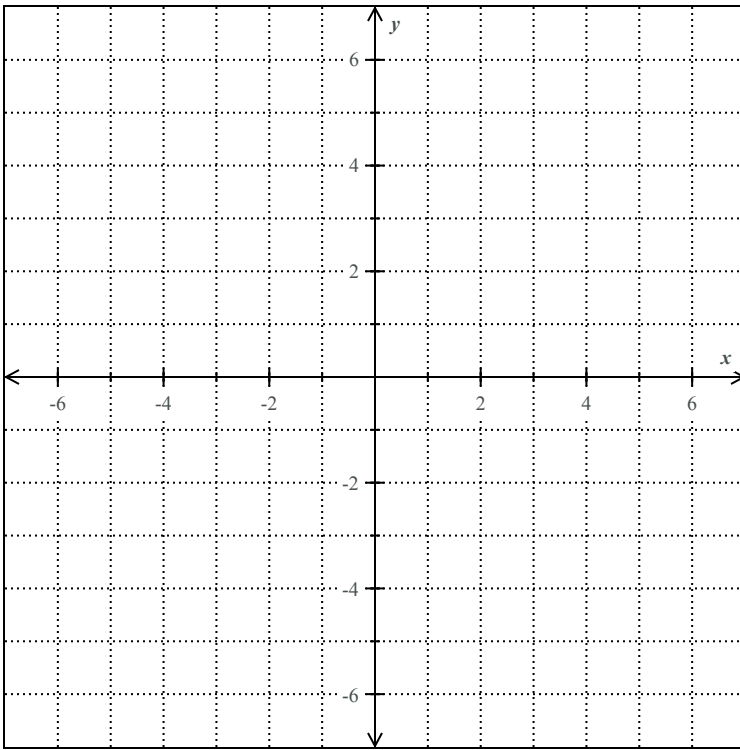
Zero(s) where the graph touches, but does not cross the x -axis:

(c) Find the y -intercept of the graph of f .

y -intercept:

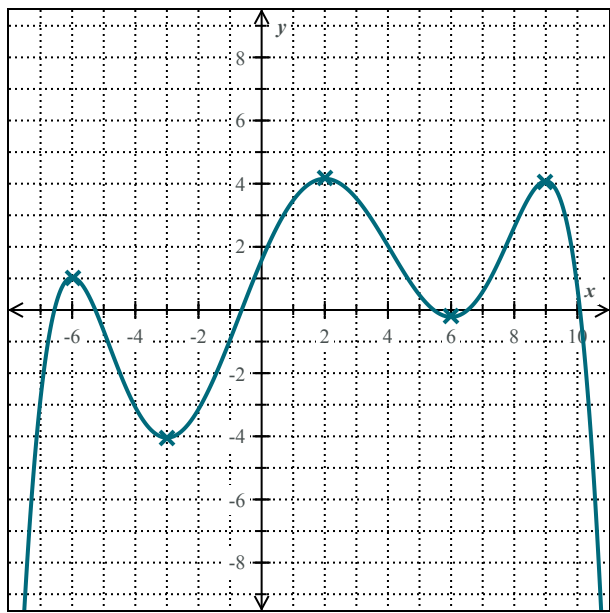
(d) Graph $f(x) = -(x + 1)^2(x - 1)$ by doing the following.

- Plot all points where the graph of f intersects the x -axis or y -axis.
- For each point on the x -axis, select the correct behavior.
- Connect the points following the behaviors.



Question 21 of 25

Below is the graph of a polynomial function with real coefficients. All local extrema of the function are shown in the graph.



Use the graph to answer the following questions.

(a) Over which intervals is the function decreasing? Choose all that apply.

☐ $(-\infty, -6)$ ☐ $(-6, -3)$ ☐ $(-3, 2)$ ☐ $(2, 6)$ ☐ $(2, 9)$ ☐ $(9, \infty)$

(b) At which x -values does the function have local minima? If there is more than one value, separate them with commas.

(c) What is the sign of the function's leading coefficient?

Positive Negative Not enough information

(d) Which of the following is a possibility for the degree of the function? Choose all that apply.

☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9

Question 22 of 25

Use the rational zeros theorem to list all possible rational zeros of the following.

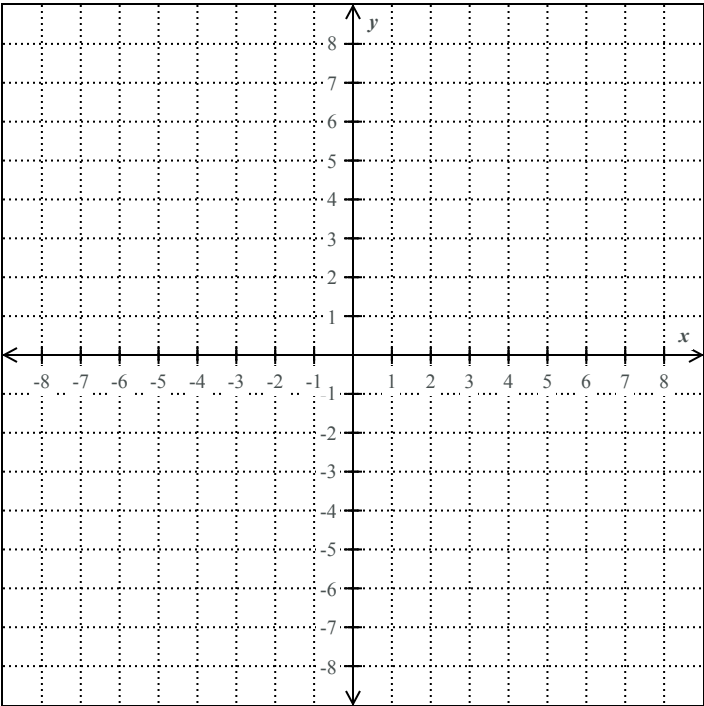
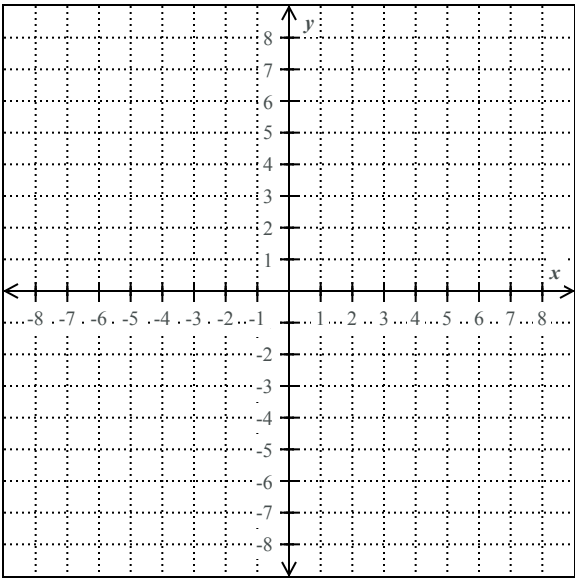
$$h(x) = 5x^3 + 5x^2 - x + 2$$

Be sure that no value in your list appears more than once.

Question 23 of 25

Graph the rational function.

$$f(x) = \frac{4x + 6}{-x - 2}$$



Question 24 of 25

Solve the following inequality.

$$x^3 - x^2 \leq -9x + 9$$

Write your answer as an interval or union of intervals.

Question 25 of 25

Class Name: **College Algebra FA26 - 014 MW**
2:30

Number of Questions: **25**

Question 1 of 25

Distance: $\sqrt{18}$

The answer can also be written in the simplified form $3\sqrt{2}$.

Question 2 of 25

$$y = \frac{5}{2}x + \frac{7}{2}$$

Note: Other correct answers include $y + 4 = \frac{5}{2}(x + 3)$ and $5x - 2y = -7$.

Question 3 of 25

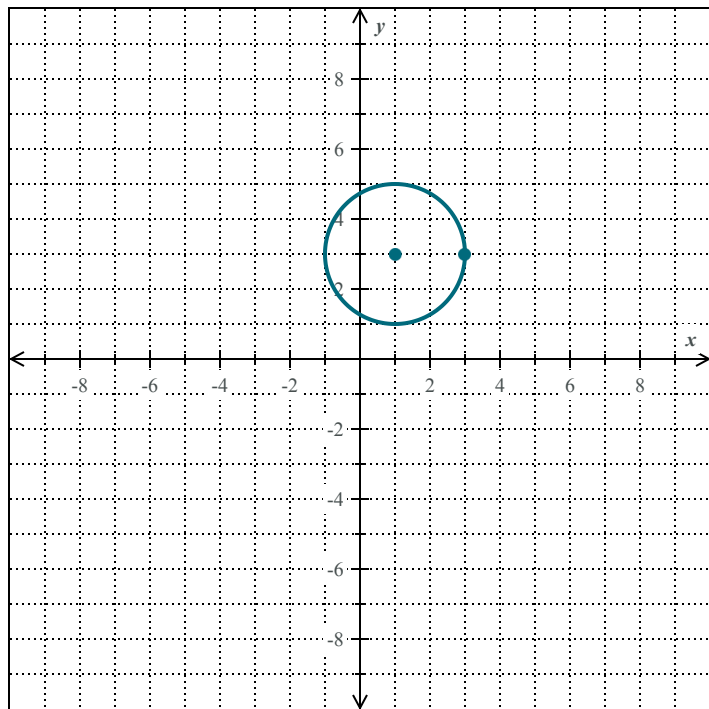
$$y\text{-intercept(s): } \frac{5}{7}$$

$$x\text{-intercept(s): } -\frac{5}{9}$$

Question 4 of 25

Center: $(1, 3)$

Radius: 2



Question 5 of 25

Domain of f : $(-\infty, -3) \cup (-3, \infty)$

Domain of g : $(-\infty, -1) \cup (-1, 6) \cup (6, \infty)$

Question 6 of 25

$$f(-3) = 2$$

$$g(8) = 6$$

$$h(-3) = \frac{16}{27}$$

Question 7 of 25

$$h(-1) = -1$$

$$h(0.5) = 1$$

$$h(2) = 2$$

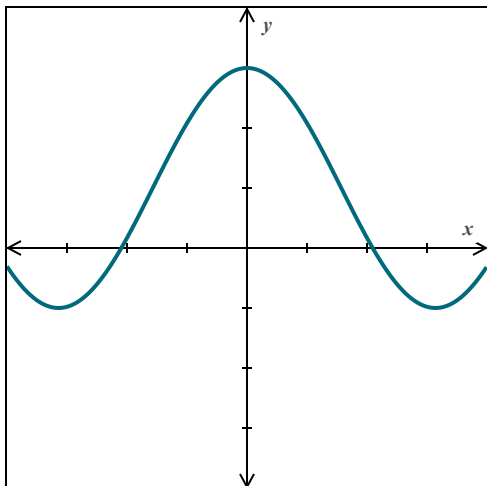
Question 8 of 25

Equation of parallel line: $y = 4x - 18$

Equation of perpendicular line: $y = -\frac{1}{4}x + \frac{15}{2}$

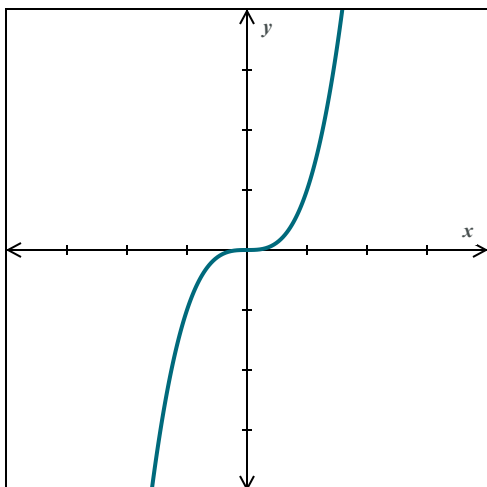
Question 9 of 25

The function r



- Even

The function s



- Odd

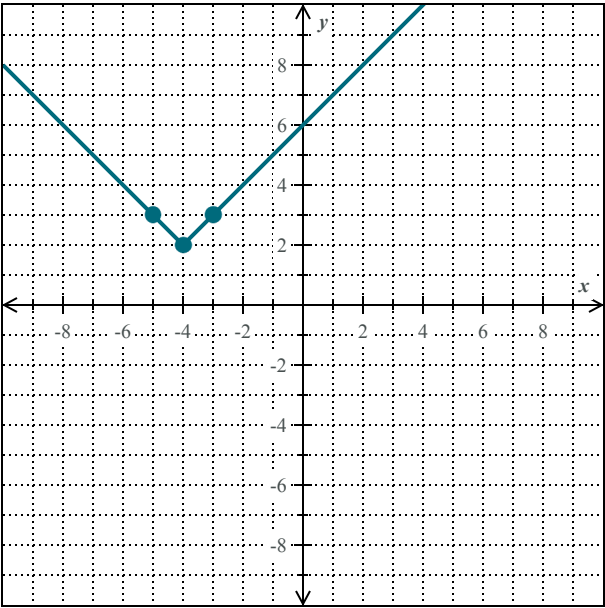
$$g(x) = -6x^5 + 7x^2$$

- Neither

$$h(x) = -4x^5 + 7x^3$$

- Odd

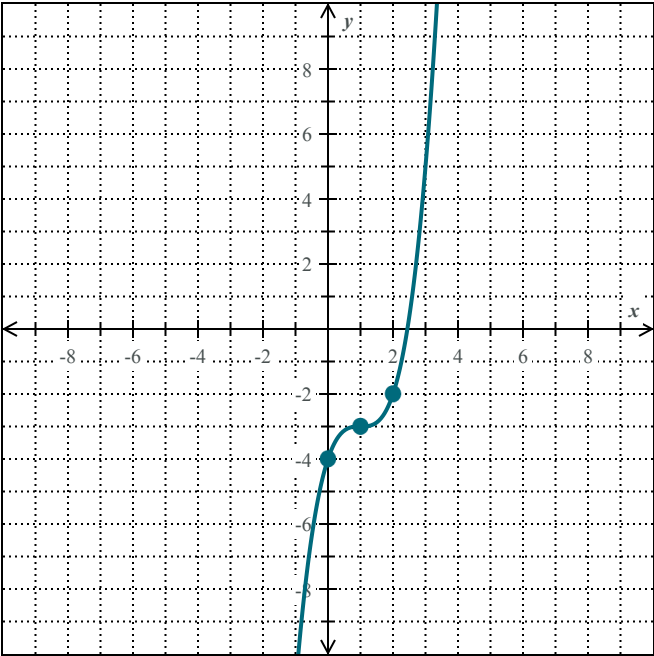
Question 10 of 25



Question 11 of 25

$g(x) = x^2 + 5$

Question 12 of 25



Question 13 of 25

$$\frac{f(x+h) - f(x)}{h} = 2$$

Question 14 of 25

$$T = 650 + 640N$$

Question 15 of 25

$$(g + h)(x) = 4x^3 + x^2$$

$$(g \cdot h)(x) = 4x^5$$

$$(g - h)(2) = 28$$

Question 16 of 25

$$(f \circ g)(x) = \frac{4x + 7}{4x + 8}$$

Domain of $f \circ g$: $(-\infty, -2) \cup (-2, \infty)$

Question 17 of 25

$$f(x) = x^2$$

$$g(x) = 8 - 3x$$

Question 18 of 25

$$M(t) = \frac{4}{3} \pi (7t + 3)^3$$

Question 19 of 25

$$g(x) = (x - 3)(x - (3 + \sqrt{7}))(x - (3 - \sqrt{7}))$$

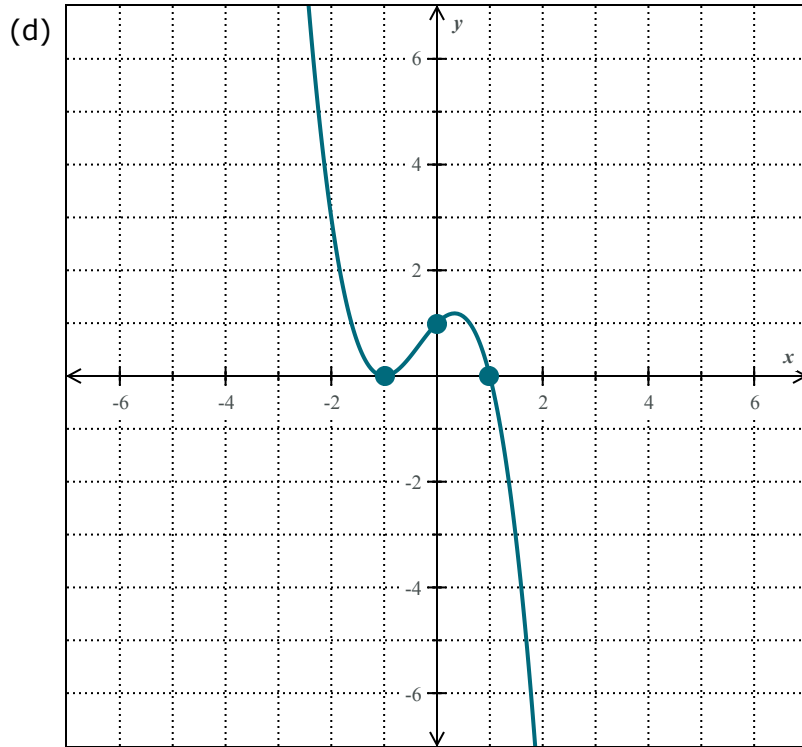
Question 20 of 25

(a) Rises to the left and falls to the right ▼

(b) Zero(s) where the graph crosses the x -axis: **1**

Zero(s) where the graph touches, but does not cross the x -axis: **-1**

(c) y -intercept: **1**



Question 21 of 25

(a) Over which intervals is the function decreasing? Choose all that apply.

☐ $(-\infty, -6)$ ☒ $(-6, -3)$ ☐ $(-3, 2)$ ☒ $(2, 6)$ ☐ $(2, 9)$ ☒ $(9, \infty)$

(b) At which x -values does the function have local minima? If there is more than one value, separate them with commas.

$-3, 6$

(c) What is the sign of the function's leading coefficient?

Negative

(d) Which of the following is a possibility for the degree of the function? Choose all that apply.

☐ 4 ☐ 5 ☒ 6 ☐ 7 ☒ 8 ☐ 9

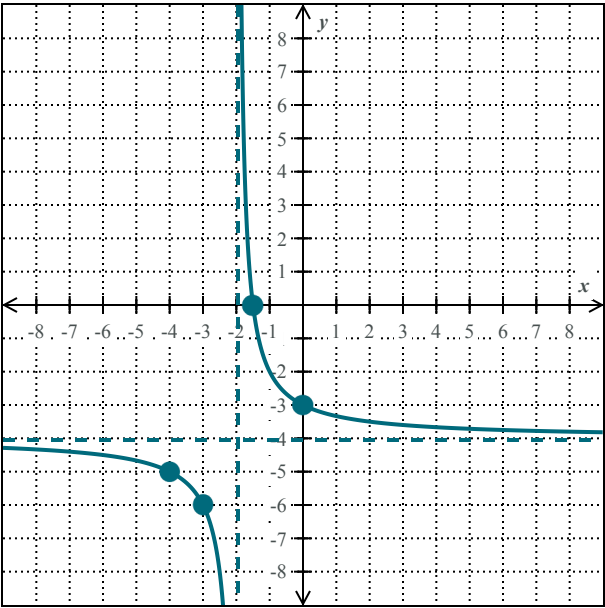
Question 22 of 25

$1, -1, 2, -2, \frac{1}{5}, -\frac{1}{5}, \frac{2}{5}, -\frac{2}{5}$

or in shorthand

$\pm 1, \pm 2, \pm \frac{1}{5}, \pm \frac{2}{5}$

Question 23 of 25



Question 24 of 25

$(-\infty, 1]$

Question 25 of 25

$(-\infty, -1] \cup (3, \infty)$